Implementing and training a diffusion-style drafter

From scratch: learnings

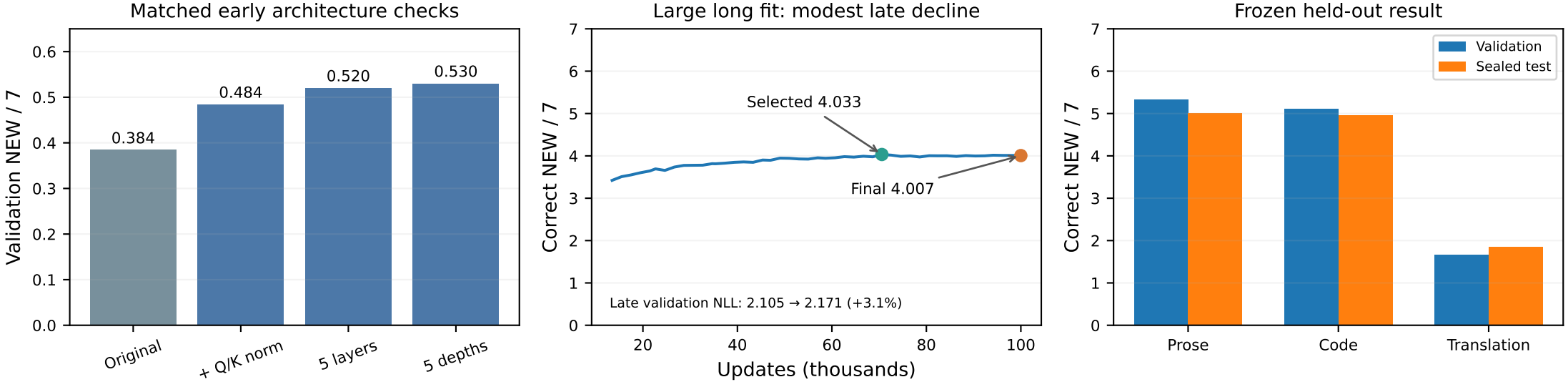
We trained a separate 79M-parameter, five-layer drafter from random weights for frozen Qwen2.5-0.5B. Following DFlash, it attends to five target hidden-state depths and predicts seven masked future tokens jointly after one known anchor. Only the drafter learns; target embeddings and vocabulary weights stay frozen.

What helped. Controlled early comparisons improved validation consecutive-token matches from 0.384 → 0.484 with query/key RMSNorm, then → 0.520 with five draft layers. Adding five feature depths reached 0.530, but that incremental benefit was uncertain. Correct target-generated labels, causal context masks, normalized conditioning, and verified generation made the pipeline credible; these implementation checks were not independent improvement ablations.

Data and response coverage mattered most. A matched 6K → 24K response study improved acceptance 0.57 → 1.02; a later 24K → 49K study reached 1.46. Longer teacher responses gave the largest subsequent gain: at 6K updates on the same 49K examples, 64 → 256-token caps improved long-context validation 1.76 → 2.96; short-context gains were smaller. The final large fits started fresh on 131K prose/code examples, then added translation to reach 196K, preserving optimizer state. Translation improved rapidly after inclusion. Multiple prose prefixes increased contexts, not independent articles.

Overfitting and unsuccessful approaches. Small datasets fit almost perfectly but generalized poorly, consistent with memorization. Larger training sets and longer teacher responses substantially reduced that problem. The large long-response run showed only modest late degradation: validation matches fell 4.033 → 4.007, while loss rose 2.105 → 2.171 after the selected checkpoint. Increasing data improved generalization; it did not eliminate overfitting. Earlier weak-decoder tests found no consistent benefit from layer dropout/null memory, probe-selected layers, or compressed Jacobian-lens conditioning. That conditioning was not the paper's sparse J-space projection; these controls still need repeating on the capable drafter.

Outcome. The selected 70.6K-update checkpoint scores 4.03/7 on validation and 3.94/7 on the once-opened test: prose 5.01, code 4.96, translation 1.84; 189 usable prompts. All 24 longer rollouts matched Qwen exactly. Actual accepted draft tokens per verification were 4.51/4.30/1.20 by domain. These are one-seed target-imitation results; repetitive teacher outputs and changing data/exposure limit causal claims. Human answer quality and speedup were not established.



One seed; different experiments have different exposure budgets. Middle/right use the same metric and scale. Matching Qwen is not human quality.

Sources: DFlash arXiv:2602.06036; Anthropic workspace/Jacobian lens; full reports and protocol in the repository.